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Riyad Babayev

EDUCATION

Bachelor of Science in **Aerospace Engineering & Computer Science** *Expected graduation date: May 2026*
University of Illinois at Urbana-Champaign (UIUC)
Honors: *James Scholar* (**GPA: 3.6/4.0**)
Relevant Courses taken**:** Engineering Materials & Structures, CAD, Flight Mechanics, Compressible/Incompressible Flow, Numerical Methods, Control Systems, Propulsion, Robotics, Aerodynamics

FEATURED PROJECTS

Racing Quadrotor Drone

- Developed and simulated real-time control algorithms (LQR + feedforward) in **Python** for high-speed drone racing
- Leveraged **NumPy, Matplotlib, and SymPy** for numerical computation, modeling, and data visualization
- Designed data-driven tuning scripts to ensure rapid response stability under maneuvers, *reducing response lag by 20%*

Spacecraft Attitude Control with Stars Tracking

- Modeled and linearized **spacecraft dynamics** with reaction wheels
- Designed a feedback control system using data from a star tracker
- Focused on precise attitude regulation in space environment

WORK EXPERIENCE

INTERNSHIP: **Audubon Companies (Engineering Consulting Company), Houston, USA** *Summer 2025 & 2024*
Field of work: **Mechanical Engineering & Software Automation**

- Designed and 3D printed *a 24" Ethane Trap Skid* for client demonstration and integration
- Implemented *automated data formatting* pipelines to eliminate repetitive manual entry, *reducing processing time by 60%*
- Utilized **AutoCAD** for modeling and drafting mechanical components and crafting the P&IDs
- Integrated *document management systems* using conditional programming logic to *interconnect data* between engineering, procurement, and compliance teams
- Drafted mechanical engineering lists and drawings, referencing P&IDs and Pipe Specifications
- Engineered *pipe and valve specifications* for a CO_2 service project
- Created detailed Material Specification Datasheets (MSDs) for piping components
- Pursued *wall thickness calculations* along with stiffness using **Nozzle PRO**
- Conducted *pipe stress analysis* using **CAESAR CX II** in accordance with ASME standards

Field of work: **Project Management**

- Supplied clients with weekly updates, reports, RFIs, ensuring information was current and accurate
- Controlled schedules and deliverables for activities for $52.7M refinery development project
- Managed a blast impact study for refinery reactors in compliance with laws, addressing proximity to two civilian buildings
- Collaborated with 7 vendors to meet client requirements, ensuring timely delivery and integration of critical components

INTERNSHIP: **AZAL (Azerbaijani Airlines), Azerbaijan** *June – August 2023*

- Consistent daily checks and monitoring of technical parameters (Diagnostics) of aircraft
- Fulfilling requirements on MOE and CAME (Continuing Airworthiness Management Exposition)
- Analyzed over 107 repetitive defects in the fleet and monitored them to prevent future complication
- Pursuing **on-site aircraft inspection** and reporting defects and dents

STUDENT ORGANIZATIONS INVOLVEMENT

ORGANIZATION: **ASAUIUC (Azerbaijan Student Association in UIUC)** *August 2024 - Current*
POSITION: **President**

- **Founded** the first registered Azerbaijani Student Organization at UIUC, aiding in cultural exchange and community building
- Expanded membership from 5 to 25 members in one semester by organizing meeting, cultural events, and guest lectures
- Secured $3,500 in event funding through successful Student Organization Resource Fee (SORF) grant applications

ORGANIZATION: **GHOST Electric Motorcycles** *October 2024 – January 2025*
POSITION: **Mechanical Engineer**

- Transitioned the team's design workflow to AutoCAD, improving design precision and reducing drafting errors by 9%, leading to faster prototyping and iteration
- Led a team of 18 members to design and improve an electric motorcycle, resulting in a 20% increase in efficiency
- Conducted tests and data analysis on 5+ motor configurations, optimizing torque output and increasing top speed
- Coordinated with suppliers to source parts, achieving a reduction in material costs for the project

SKILLS

Software

- **Programming**: Java, Python, C++
- **CAD Software**: NX Siemens, AutoCAD, SolidWorks, Navisworks
- **FEA & Analysis tools**: Cura, ANSYS, ABAQUS, Nozzle PRO, CAESAR CX II
- **Microsoft Office**: Word, Excel, PowerPoint | Bluebeam Revu

Languages

- English, Azerbaijani, Russian, Turkish